PICASSO
Robotic Part - Conceptual Design

Draft 0.1

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1. Overall view

The following drawing shows the robot in operation, the surfaces where the robot goes (in grey), and different positions of the ionisation chamber (on the circle, inside the circle and close to the bed). Localisation beacons appear on the wall.

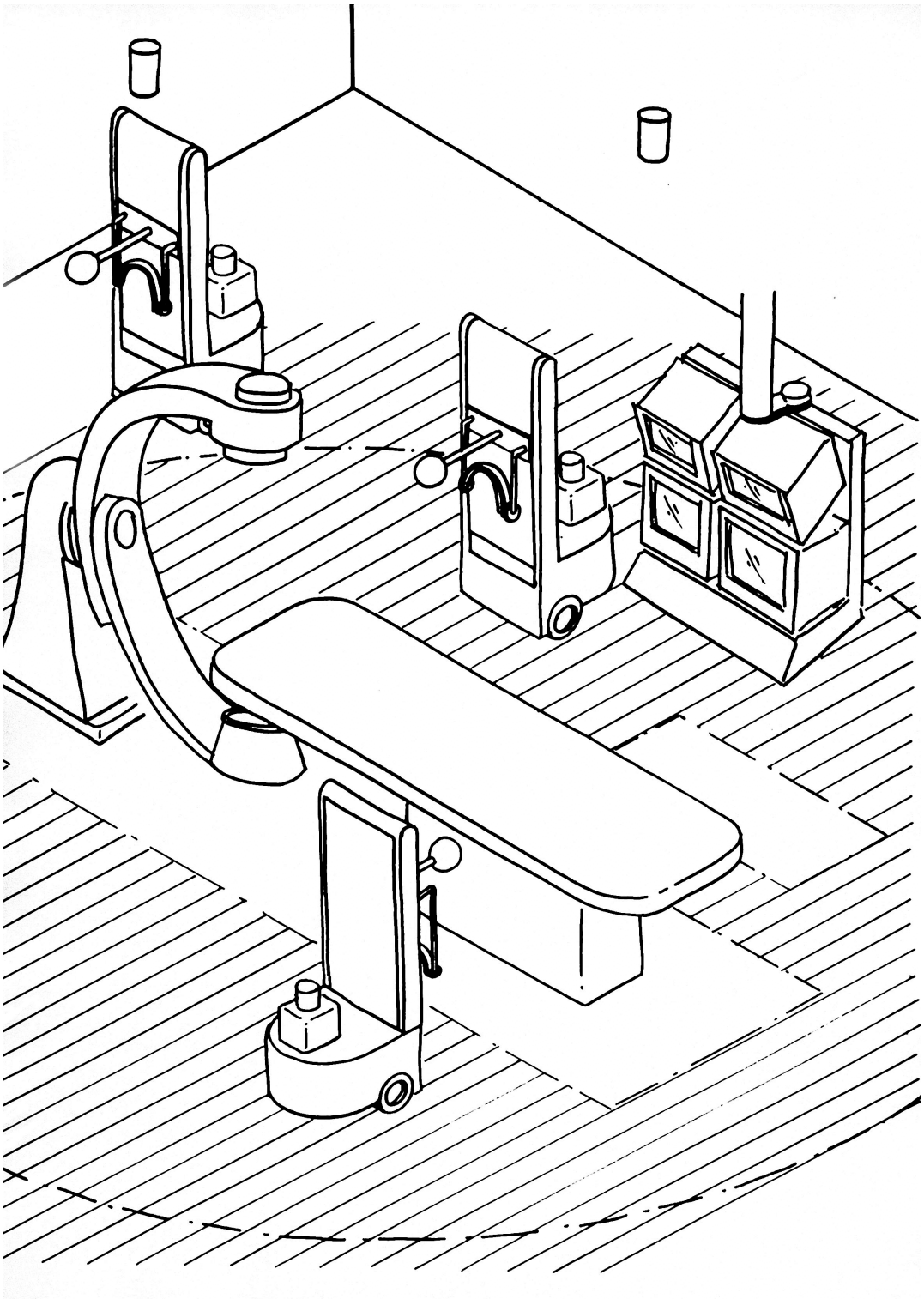


Fig 1 : The robot in different measuring positions

2. Mechanical Design

2.1. The robot

This is a robot based upon the Robulab platform (see attachment 1), on top of which is installed the lift mechanism.

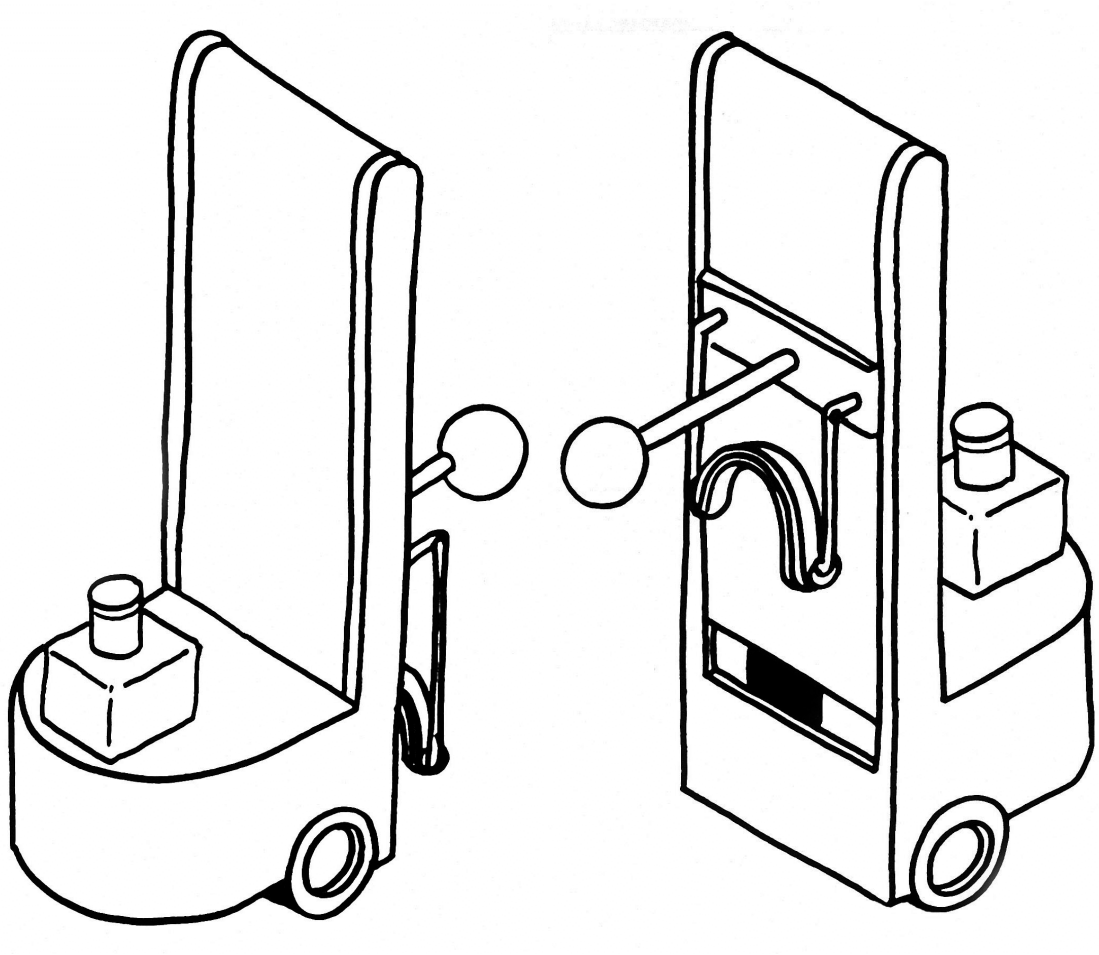


Fig. 2 : Front and rear views of the robot

2.2. The lift mechanism

It is a replication of the system we use in Oscar refueling robot. It allows the ionisation chamber to move from 50 cm to 150 cm.

2.3. Protection of the Ionisation chamber

We propose a simple mechanism which lifts and opens with 2 motors when the robot is moving.

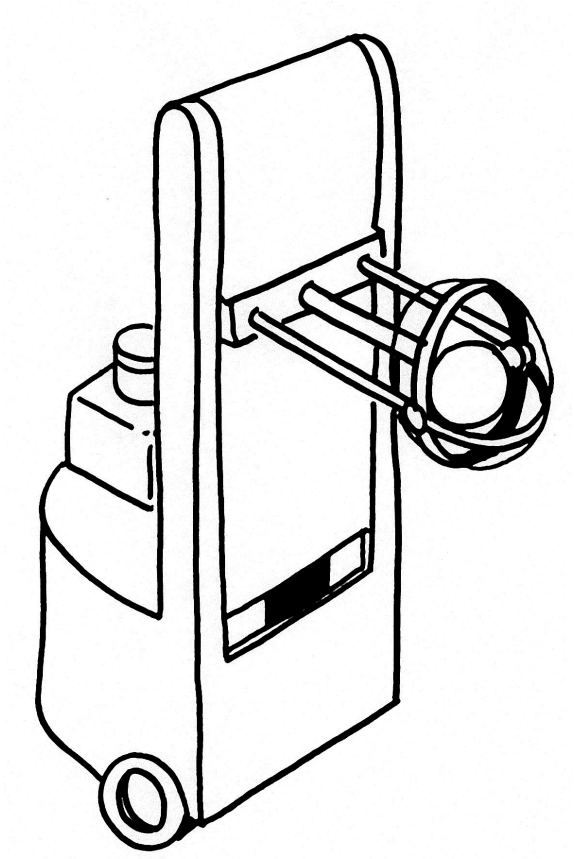


Fig. 3 : Mechanical protection of the ionisation chamber (only when moving)

3. Localization

The coordinates X,Y and Z are given into a reference frame linked to the table stand. The odometry attached to the wheels will be used to control position from one location to another one, but absolute localization is needed at measuring positions. We propsoe to use a rotating laser with reflective beacons instead of the initial map matching method which did not lead to satisfying results (because of the angle of the laser, see attachment 5). The room has to be equipped with a set a beacons able to be detected using a full rotating laser scanner. The passive beacons consist of reflecting cylinders which positions are known into the absolute reference frame linked to the scanner room. The rotating laser scanner sweeps the area in front of it, hence detects the signals returned by each beacons and then record each beacon angular position expressed into the robot local reference frame. The angular information can then be used to determine (mathematically by triangulation) the position of the robot. The advantage of this arrangement is that the position and heading can be determined from a single robot position (ie. it does not rely on robot movement to calculate heading). Also, this technique does not use the conventional method of building up a complex map of its environment to determine its position. It is only needed to know the placement of each beacon.
Such devices are devices scanning around in a plane and "actively" measuring the distances and angles to reflectors mounted in this plane.The system already knows the position of reflectors in a "world coordinate system" and uses these to continually calculate its own position and orientation which are then ready for transmission.

Details of 2 rotating scanners are given in attachments 2 and 3.
An example of beacons placement is given by fig 4.

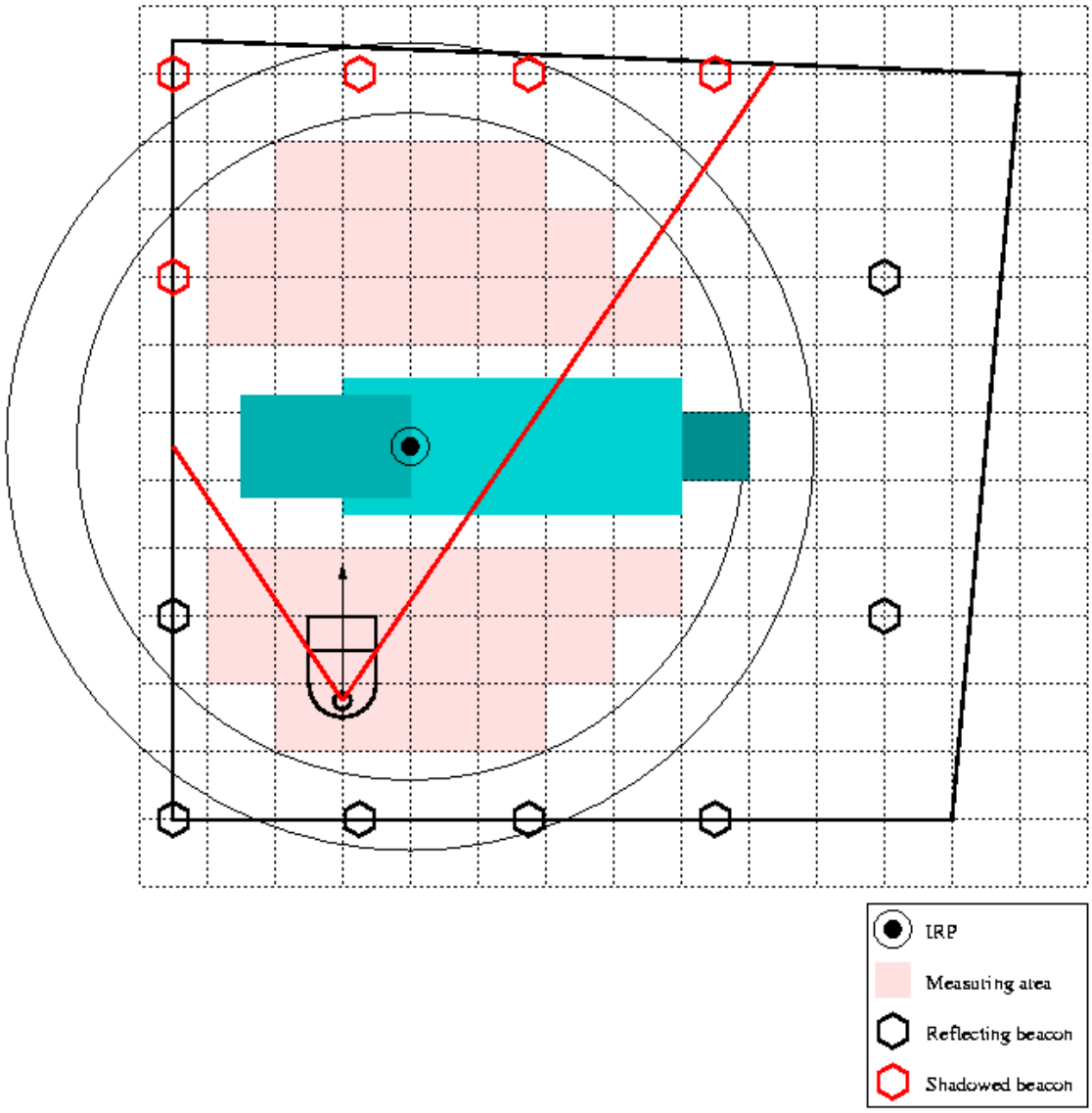


Fig. 4 : An example of beacons placement

4. Control system

4.1. Control system

The lift mechanism and motions of the robot are controlled by an on-board micro-controller based on the MPC555 from Motorola. Communication is done through a wireless serial line. A description of the robot controller is given in attachment 4.

4.2. Control primitives

Each acquired data is made of a value returned by a ionization chamber and the space coordinates of the ionization chamber position (X, Y and Z).

According to the straight line strategy (the robot only moves backwards and forwards, and skips from one straight line to another), the following basic commands are needed :

- MOTN = pure translations and rotations of the robot [available]
- JSTK = manual control with the joystick [available]
- XYZT = returns the position estimated by the localization system [to be developed]

5. Contribution to use case

There is an intialization process where beacons need to be installed, and the internal map built by the system. To make easier this process, we suggest installing a beacon just underneath the IRP point (which may be removed when operating measurements).

6. Attachments

1. RobuLab Pro platform specifications
2. Robosense specifications
3. IBEO LADAR 2D NAV localisation laser specifications
4. Specifications of the RSMPC555
5. Localization tests with the SICK laser